



13.4.4 Wrap-Up: Reflection

Draw an **X** on each line to indicate how you feel about each learning target.

1. I can match multiple representations of the parent function to its table of values.

I need help.	I'm getting there, but	I understand this
I can't get started.	I need more practice.	and I feel confident.



2. I can determine a function type that could represent a transformation of a parent function.

I need help.	I'm getting there, but	I understand this
I can't get started.	I need more practice.	and I feel confident.



Unit 13, Lesson 5: Exploring Transformations – Horizontal Shifts



Benchmark

MA.912.F.2.1 Identify the effect on the graph or table of a given function after replacing $f(x)$ by $f(x) + k$, $kf(x)$, $f(kx)$, and $f(x + k)$ for specific values of k .



Learning Targets

- ☐ I can discover the effect on the graph of $f(x)$ for a transformation of $f(x + k)$.
- ☐ I can discover the effect on a table of values of $f(x)$ for a transformation $f(x + k)$.



13.5.1 Warm-Up: Megaships

1. Florida has 15 seaports.

A seaport is where ships can dock to load and unload cargo and passengers. In 2016, the Port of Miami was updated to have a depth of 50 feet so that megaships could dock there. A megaship can hold 406 containers. A normal ship can hold 157 containers.



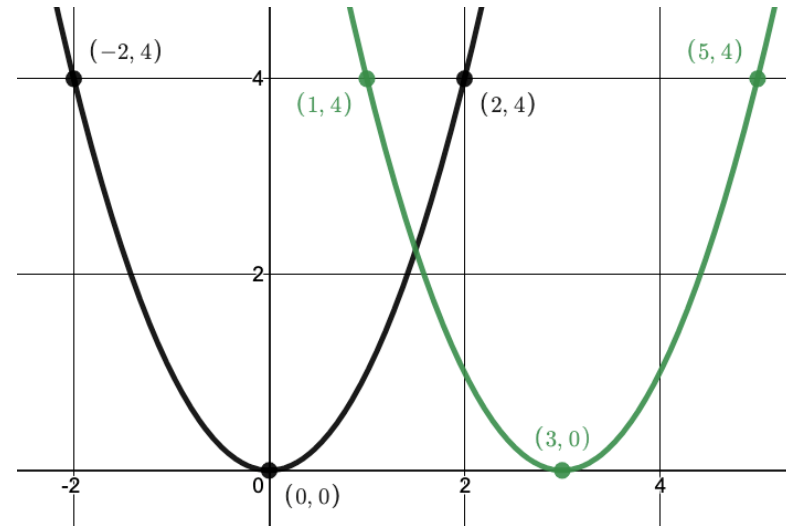
An article on *JOC.com* said, "Container terminals in gateway ports around the world, already suffering from congestion caused by mega-ships, may not be able to keep up with their ever-increasing size because of limits on crane technology and the lack of yard space."

- a. Based on the capacity of a megaship, why do you think there could be an issue with the lack of yard space?
- b. A standard shipping container can hold about 24,900 tin cans. How many more tin cans can be delivered by a megaship than by a normal ship?



13.5.2 Exploration: Shifting the Graph of a Function

1. The graphs of $y = x^2$ and $y = (x - 3)^2$ are shown.



- a. What do you notice about the graphs of $y = x^2$ and $y = (x - 3)^2$?
- b. Work with your partner to describe how to shift the graph of $y = x^2$ so that it coincides with the graph of $y = (x - 3)^2$.

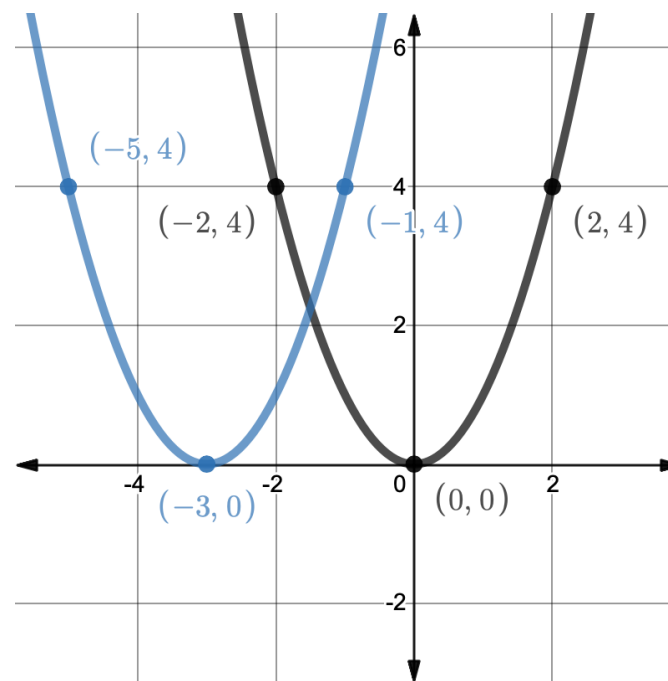
- c. The table of values for $y = x^2$ and $y = (x - 3)^2$ are shown.

$y = x^2$		$y = (x - 3)^2$	
x	y	x	y
-2	4	1	4
0	0	3	0
2	4	5	4

What do you notice when comparing the table of values?

- d. Discuss with your partner connections between the x -values in the table of values and the description of shifting the graph of $y = x^2$ so that it coincides with the graph of $y = (x - 3)^2$. Summarize your discussion.

2. The graphs of $y = x^2$ and $y = (x + 3)^2$ are shown.



- a. Work with your partner to describe how to shift the graph of $y = x^2$ so that it coincides with the graph of $y = (x + 3)^2$.
- b. The table of values for $y = x^2$ and $y = (x + 3)^2$ is shown.

$y = x^2$		$y = (x + 3)^2$	
x	y	x	y
-2	4	-5	4
0	0	-3	0
2	4	-1	4

Discuss with your partner connections between the x -values in the table of values and the description of shifting the graph of $y = x^2$ so that it coincides with the graph of $y = (x + 3)^2$. Summarize your discussion.

c. Complete the statements.

The graph of $y = (x + 3)^2$ is a
☐ horizontal shift
☐ vertical shift

of the graph of $y = x^2$.

For the graph of $y = x^2$ to coincide with the graph of

$y = (x + 3)^2$, it is shifted
☐ down
☐ left
☐ right
☐ up
 3 units.

The shift is seen in the table of values by

☐ adding
☐ subtracting
 to each
☐ x -value
☐ y -value

for each corresponding
☐ x -value.
☐ y -value.



13.5.3 Guided Instruction: Writing Transformations Using Function Notation

The shifts of the graphs can also be written in function notation. The original graph $y = x^2$ can be written as $f(x) = x^2$.

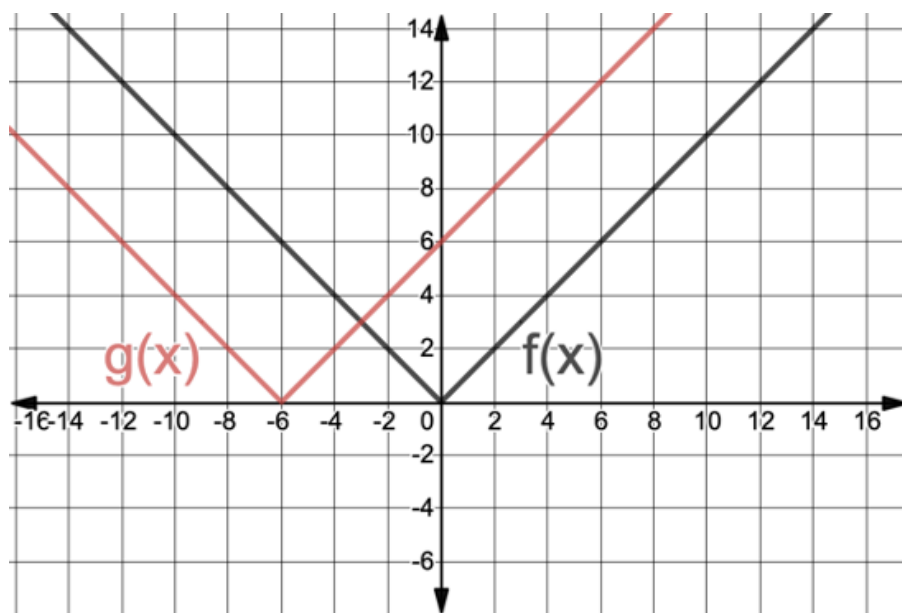
- Notice that the horizontal shift to the right 3 is written as $y = (x - 3)^2$, where the $(x - 3)$ is in place of the x . To write this shift using function notation, substitute $x - 3$ in place of x .

$$\begin{array}{c}
 f(x) = x^2 \\
 \swarrow \searrow \\
 \boxed{x - 3} \\
 f(\boxed{}) = (\boxed{})^2
 \end{array}$$

- Work with your partner to complete the table. The first row has been completed.

Original Equation	Original Function	Shift	Transformed Function
$y = x^3$	$f(x) = x^3$	Right 5	$f(x - 5) = (x - 5)^3$
$y = x^2$		Left 7	
$y = \sqrt{x}$		Right 2	
$y = 3^x$		Left 1	

b. The graphs of $y = f(x)$ and $y = g(x)$ are shown.



How does the graph of $f(x)$ compare to the graph of $g(x)$?

c. Two function descriptions are shown. Discuss with your partner which function description matches the transformation on the graph. Then, justify your choice.

- $g(x) = f(x + 6)$
- $g(x) = f(x - 6)$

Justification

2. The table of values for the function $y = B(x)$ is shown.

x	B
-4	15
-1	8
5	-6
14	-27

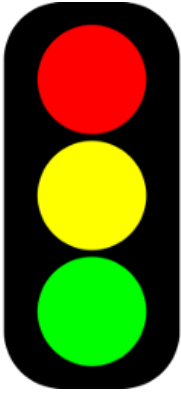
- a. Describe the effect that the transformation of $B(x - 2)$ will have on the x -values for each corresponding B -value in the table.
- b. Describe the effect that the transformation of $B(x + 7)$ will have on the x -values for each corresponding B -value in the table.



13.5.4 Wrap-Up: Reflection

1. Describe your learning today by writing a short statement for each.

Stop, I'm confused about ...



Proceed with caution,
I need clarification on ...

Go ahead. I understand ...

Unit 13, Lesson 6: Exploring Transformations – Vertical Shifts



Benchmark

MA.912.F.2.1 Identify the effect on the graph or table of a given function after replacing $f(x)$ by $f(x) + k$, $kf(x)$, $f(kx)$, and $f(x + k)$ for specific values of k .



Learning Targets

- ☐ I can discover the effect on the graph of $f(x)$ for a transformation of $f(x + k)$ or $f(x) + k$.
- ☐ I can discover the effect on a table of values of $f(x)$ for a transformation of $f(x + k)$ or $f(x) + k$.